

Victor Hugo
Mosquera Alvarado

SENIOR FULLSTACK WEB & WEB3

Senior Full-Stack & Web3 developer with 6+ years building scalable APIs, microservices, and decentralized systems. I apply hexagonal architecture,DDD, and SDD (Spec-Driven Development) to deliver clean, well-tested, production-ready code. Active user of AI tools throughout the development lifecycle.

EXPERIENCE

Backend Developer — Puntos Colombia (PCO)

10/2025 – present

XpertGroup · Colombia

Stack: Node.js 22 · TypeScript · Fastify 5 · NestJS · Awilix · Prisma · PostgreSQL · DynamoDB · AWS · Azure DevOps · Docker · Hexagonal Arch · DDD · CQRS

400+ commits250+ pull requests9 repositories

- Led backend development of the **users microservice** (pco-app-users-back) applying hexagonal architecture,DDD and CQRS — session & login, logout with **Cognito token revocation via STS role assumption**, device limit control, user deactivation, and member device status.
- Built **user segmentation** integrated with MLOps models, querying DynamoDB in batch (findByIds), and a **product recommendations endpoint** configured via AWS SSM Parameter Store.
- Integrated external systems: **Marketplace (MKP)** for sale orders, **Travel API** with custom auth, **Adjoe** for gamification payouts, **Bancolombia** account linking, and **POS** with OAuth token auth.
- Hardened security: **signature validation** on sensitive endpoints, XSRF handling, encryption with corporate library (pco-crypto), sensitive data masking in logs, and strengthened authorization middleware.
- Improved observability: request traceability, memory usage endpoint, DB health check with cache, and fixed singleton/scoped DI lifecycle bugs — verified with tests.
- Developed **digital wallet features** (PaaS Wallet / NestJS): JWT claims middleware & guard to prevent cross-user data manipulation; payment flows with **Wompi and Bancolombia Pay**; DTO validation with class-validator (422 responses); and notification service integration.
- Migrated **10+ Lambda functions** (pco-monorepo-apigw-aws) from CommonJS to ES Modules, Jest to Vitest, and Node.js 18 to **Node.js 22**. Migrated API Gateway to v2. Currently migrating email service from HTTP to **AWS EventBridge**.
- Managed multi-environment configuration (dev/QA/prod) in **SSM Parameter Store** and **Serverless Framework** across MLOps, device limits, MKP, travel, sessions, and email/SMS templates.
- Documented APIs with **OpenAPI/Swagger**, wrote unit tests with Vitest, and maintained Conventional Commits + Git Flow by sprints.

Full-Stack Web & Web3 Developer

09/2019 – 06/2025

Evolution Systems B.V. · Apeldoorn, Netherlands (remote)

Stack: Node.js · Angular · React · XRPL · Solana · EVM · NestJS · Microservices · Docker

- Built a **cross-chain DEX** for CSC tokens (XRPL → ETH, BSC, SOL, AVAX) with admin panel, public API gateway, and per-chain microservices.
- Developed a **PWA NFT marketplace** on XRPL and a **P2P betting PWA** on XRP Ledger + Solana integrated with Telegram Bot and MiniApp.
- Contributed to a **decentralized poker dApp** (Xumm) and **TrustTracker™** — real-time casino spins & bets on XRP Ledger.
- Built desktop/mobile wallets for CasinoCoin and Crazy8Token; roulette interface for Cammegh.
- API documentation (Swagger), Linux server administration, bug fixing and research.

NOTABLE PROJECTS

NFT Marketplace dApp

Next.jswagmiSolidityFoundry

Lottery Smart Contract

SolidityFoundryChainlink

SPL Token 2022 (Solana)

TypeScriptsolana/web3.js

Stablecoin dApp

Next.jsviemSolidityFoundry

MyIVO — Invoice App

NestJSHexagonalSDDDIAN

ERC-20 / ERC-721 tokens

SolidityAvalancheCelo

BACKEND

Node.jsTypeScriptNestJSFastify 5
Express.jsPrismaTypeORMMicroservices
Hexagonal ArchDDDCQRSSTDTDD
WebSocketsJWT / OAuth2Swagger

AWS

LambdaAPI Gateway v2CognitoSQSSNS
DynamoDBSSM Param StoreEventBridgeSTS
S3KMSSecrets ManagerEC2
Serverless Framework

FRONTEND

AngularReactNext.jsIonic PWA
TailwindRxJSFlutterTelegram MiniApps

DATABASES

PostgreSQLDynamoDBMongoDBMySQLRedis

DEVOPS & TOOLS

DockerAzure DevOpsGit FlowCI/CD
SonarQube

WEB3 & BLOCKCHAIN

XRPLSolanaEVMSolidityFoundry
WagmiViemEthers.jsTheGraphDeFi

AI & TOOLS

ClaudeGitHub CopilotMCP (TypeScript SDK)
Cursor AILLM integration

EDUCATION

Systems Engineering

Universidad de los Llanos
2017 · Villavicencio, Colombia

CERTIFICATIONS

IBM AI Engineering Professional	2025
• In progress — Coursera	
Decentralized Finance (DeFi) — Duke University	2024
Solidity 101 — Cyfrin Updraft	2024
Flutter Zero to Expert — Udemy	2023
R Programming — Coursera	2023

CURRENTLY LEARNING

PythonRustAWS Cloud Arch. Professional
AI Engineering